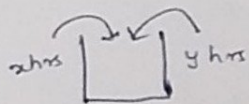


PIPES AND CISTERNS

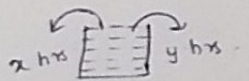
1. There are 2 types of pipes,

* inlet pipe $\rightarrow$ pipe which fills the tank or cistern

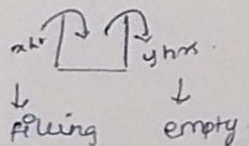
* outlet pipe $\rightarrow$ any pipe which empties the tank
/ leakage



together means to fill tank in $\frac{xy}{x+y}$ hrs. (Inlet pipes)



working together, tank emptied in $\frac{xy}{x+y}$ hrs (outlet)



(i) $x < y$ tank is always filled.

Tank is filled in $\frac{xy}{y-x}$ hrs.

(ii) $x > y$ Tank is always emptied.

Tank is emptied in $\frac{xy}{x-y}$ hrs.

1). A pipe can fill a tank with water in 2 hrs. Because of a leak, it took $2\frac{2}{3}$ hrs to fill the tank. In how many hrs, the leak can be empty the full tank?

Ans: 8

A $\xrightarrow{\text{fill}}$ 2 hrs
inlet

$x > 2$

$$\frac{2x}{x-2} = \frac{8}{3}$$

B $\longrightarrow$ x hrs.
outlet

$$6x = 8x - 16$$

$$x = 8$$

2) Two pipes A & B together can fill cisterns in 4 hrs. If they are opened separately ~~we~~^B will take 6 hrs more than A. to fill the cisterns. In how many hrs, A alone can fill the cistern separately?

Ans: 6

A $\rightarrow x$ hrs

B $\rightarrow (x+6)$ hrs

$$\frac{x(x+6)}{x+x+6} = 4$$

$$x^2 + 6x = 8x + 24$$

$$x^2 = 2x + 24$$

$$x^2 - 2x - 24 = 0$$

$$x^2 - 6x + 4x - 24 = 0$$

$$x(x-6) + 4(x-6) = 0$$

$$x = 6, (-4)x$$

$$x = 6$$

3) Two pipes A & B can fill a tank in 24 min and 32 min respectively. If both pipes are opened simultaneously, After how much time B should be closed so that the tank will be full in 18 min.

Ans: 8 min

Let pipe B is closed after x min

18 min work of A + x min work of B = 1

$$18 \times \frac{1}{24} + x \times \frac{1}{32} = 1$$

$$\frac{x}{32} = 1 - \frac{3}{4} = \frac{1}{4}$$

$$x = \frac{32}{4}$$

$$x = 8 \text{ min}$$

4) pipes A & B can fill a tank in 36 hrs & 48 hrs respectively. Both pipes are opened together for 9 hrs. And then pipe A is closed. In how much time pipe B can fill the remaining part of the tank.

Ans: 27 hrs

$$9 \text{ hrs } (A+B) = 9 \left(\frac{1}{36} + \frac{1}{48} \right) = 9 \times \frac{48+36}{36 \times 48} = \frac{7}{16} \text{ tank is filled}$$

$$\begin{array}{l} 1 \xrightarrow{B} 48 \text{ hrs} \\ \frac{9}{16} \xrightarrow{B} x \end{array} \Rightarrow \frac{48}{1} \times \frac{9}{16} = 27 \text{ hrs}$$

5) Pipe A can fill a tank of capacity 350 litres in $3\frac{1}{2}$ min, pipe B can fill a tank of capacity of 780 litres in $8\frac{2}{3}$ min. How long it will take to fill a tank with capacity 1615 litres, when both pipes are opened together.

Ans: $8\frac{1}{2}$ mins

A $\frac{7}{2} \text{ min} \rightarrow 350$

1 min $\rightarrow \frac{350 \times 2}{7} = 100 \text{ litres}$

B $\frac{26}{3} \text{ min} \rightarrow 780$

1 min $\rightarrow \frac{780 \times 3}{26} = 90 \text{ litres}$

1 min $\xrightarrow{A+B} 100+90 = 190 \text{ (together)}$

$x \rightarrow 1615$

$\frac{1}{190} \times 1615 = \frac{17}{2} = 8\frac{1}{2} \text{ mins}$

6) Two pipes A & B can fill a tank in 16 hrs & 20 hrs. They are opened alternatively for 1 hr each starting with pipe A. In how many hrs, empty tank can be filled completely.

$$\boxed{\text{Ans: } 17\frac{3}{4} \text{ hr}}$$

$$\begin{array}{c} A+B \\ (2 \text{ hrs}) \end{array} \xrightarrow{A+B} \frac{1}{16} + \frac{1}{20} = \frac{9}{80}$$

↓ 8

16 hrs →

↓ 8

$\frac{72}{80}$ completed

$$17 \text{ hrs} \xrightarrow{A} \frac{1}{16} \times \frac{5}{5} = \frac{5}{80} \quad \checkmark \quad \frac{2}{80} \text{ (Remaining)}$$

$$18 \text{ hrs} \xrightarrow{B} \frac{1}{20} \times \frac{4}{4} = \frac{4}{80} \rightarrow \frac{3}{4} \text{ hrs only.}$$

so 17 complete hrs & $\frac{3}{4}$ hrs of 18th hr.

$$\therefore 17\frac{3}{4} \text{ hr.}$$

7) Pipes A & B are filling pipes while C is emptying pipe. A & B can fill the tank in 72 mins & 90 mins respectively. When all the 3 pipes are opened together, tank is filled in 2 hrs. A & B are opened together for 12 min and close. Now C is opened, after how many mins, tank is emptied completely?

$$\boxed{\text{Ans: } 18}$$

$$A + B + C \xrightarrow{1 \text{ min}} \frac{1}{120}$$

$C \rightarrow x$ min to empty tank.

$$\frac{1}{72} + \frac{1}{90} - \frac{1}{x} = \frac{1}{120}$$

$$\frac{1}{72} + \frac{1}{90} - \frac{1}{120} = \frac{1}{x}$$

$$\frac{90+72}{72 \times 90} - \frac{1}{120} = \frac{1}{x}$$

$$\frac{8+9}{162}$$

$$\frac{72 \times 90}{364}$$

$$\frac{1}{40} - \frac{1}{120} = \frac{1}{x}$$

$$\frac{120-40}{4800} = \frac{1}{x}$$

$$\frac{80}{4800} = \frac{1}{x}$$

$$8x = 480$$

$$x = \frac{480}{8} = 60$$

$$\boxed{x = 60}$$

$$12(A+B) = 12 \left(\frac{1}{72} + \frac{1}{90} \right) = 12 \times \left(\frac{90+72}{72 \times 90} \right)$$

$$= \frac{12}{12} \times \frac{1}{40} = \frac{1}{40}$$

$$= \frac{3}{10} \text{ in } 12 \text{ min.}$$

$\hookrightarrow$ it needs to be emptied.

$$1 \xrightarrow{C} 60 \text{ min}$$

$$\frac{3}{10} \longrightarrow \frac{3}{10} \times 60$$

$$= 18 \text{ min}$$

8) Three pipes A, B, C fill a tank together in 4 hrs.

pipe C can fill a tank at twice the speed of pipe B. pipe B fill the tank at thrice the speed of pipe A. Find time taken by pipe A to fill the complete tank.

Ans: 40 hrs

$$A \rightarrow x \text{ hrs}$$

$$B \rightarrow \frac{x}{3} \text{ hrs}$$

$$C \rightarrow \frac{1}{2} \left(\frac{x}{3} \right) = \frac{x}{6} \text{ hrs}$$

$$A+B+C \rightarrow \frac{1}{4}$$

$$\frac{1}{x} + \frac{3}{x} + \frac{6}{x} = \frac{1}{4}$$

$$x = 40 \text{ hrs}$$

$$C = 2B$$

$$B = 3A$$

$$S \propto \frac{1}{t}$$

$\Rightarrow$ speed is double $\rightarrow$ time is half

$\Rightarrow$ speed is 3 times $\rightarrow$ time $\frac{1}{3}$

9) Pipes A & B can fill a tank in 16 hrs & 24 hrs. Pipe C can empty the tank in x hrs. All the pipes are opened together at 10:30 am but C was closed at 2:30 pm. If the tank was at 8:30 pm on the same day. Find the value of x .

Ans: 96

$$4 \left(\frac{1}{16} + \frac{1}{24} - \frac{1}{x} \right) + 6 \left(\frac{1}{16} + \frac{1}{24} \right) = 1$$

$$10 \left(\frac{1}{16} + \frac{1}{24} \right) = 1 + \frac{4}{x}$$

$$\frac{25}{24} = 1 + \frac{4}{x}$$

$$x = 24 \times 4$$

$$x = 96$$

'x' hrs

'C'

10:30 am

2:30 pm

4 hrs

10) Two pipes can fill the tank in 25 min & 30 min. And a waste pipe empties three gallons per min. All the three pipes are opened together & fill the tank in 15 min. Find the capacity of the tank.

Ans: 450 gallons

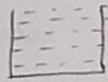
A $\rightarrow$ 25 min
B $\rightarrow$ 30 min } inlet

$$\Rightarrow \frac{1}{25} + \frac{1}{30} - \frac{1}{x} = \frac{1}{15}$$

C $\rightarrow$ $\frac{3 \text{ gallons/min}}{x \text{ min}}$

$$\Rightarrow \frac{1}{x} = \frac{6+5-10}{150} = \frac{1}{150}$$

capacity $\Rightarrow 150 \times 3 = 450$
gallons.



11) A tap can fill a tank in 4 hrs. After half of the tank is filled, 2 similar taps are opened. What is the total time taken to fill the tank.

Ans: 2 hr 40 min

$$T = 2 \text{ hr} + 40 \text{ min}$$

2 hrs $\rightarrow \frac{1}{2}$ tank is filled.

1 tap $\rightarrow \frac{1}{4}$

$\frac{3}{4} \rightarrow 60 \text{ min}$

2 taps $\rightarrow \frac{3}{4}$

$\frac{1}{2} \rightarrow ?$

$$\frac{60}{3/4} \times \frac{1}{2}$$

40 min

12) 4 pipes can fill a tank in 15, 20, 30 & 60 hrs. The 1st pipe was opened at 6 am, 2nd at 7 am, 3rd at 8 am, 4th at 9 am. At what time the tank will be filled completely.

Ans: 1 pm

$$A \rightarrow x \text{ hrs}$$

$$B \rightarrow (x-1) \text{ hrs}$$

$$C \rightarrow (x-2) \text{ hrs}$$

$$D \rightarrow (x-3) \text{ hrs}$$

$$x \text{ hrs } A + (x-1) \text{ hrs } B + (x-2) \text{ hrs } C +$$

$$(x-3) \text{ hrs } D = 1$$

$$\Rightarrow \frac{x}{15} + \frac{x-1}{20} + \frac{x-2}{30} + \frac{x-3}{40} = 1$$

$$\boxed{x=7}$$

$$\Rightarrow 6 \text{ am} + 7 \text{ hrs} = 1 \text{ pm.}$$